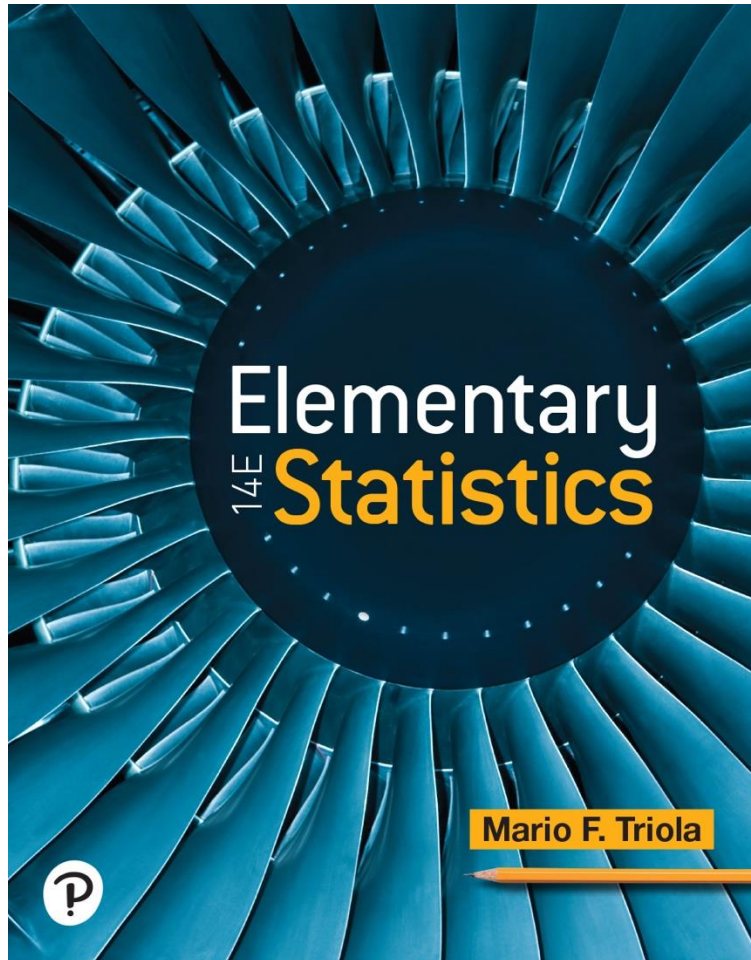


Elementary Statistics

Fourteenth Edition



Chapter 4 Probability

Probability

4-1 Basic Concepts of Probability

4-2 Addition Rule and Multiplication Rule

4-3 Complements and Conditional Probability

4-4 Counting

Key Concept

We extend the use of the multiplication rule to include the probability that among several trials, we get **at least one** of some specified event. We consider **conditional probability**: the probability of an event occurring when we have additional information that some other event has already occurred.

Complements: The Probability of “At Least One”

When finding the probability of some event occurring “at least once,” we should understand the following:

- “At least one” has the same meaning as “one or more.”
- The **complement** of getting “at least one” particular event is that you get **no** occurrences of that event.

The Probability of “At Least One” (1 of 2)

Finding the probability of getting **at least one** of some event:

1. Let A = getting **at least one** of some event.
2. Then $\bar{A}$ = getting **none** of the event being considered.

The Probability of “At Least One” (2 of 2)

3. Find $P(\bar{A})$ = probability that event A does not occur.
4. Subtract the result from 1. That is, evaluate this expression:

$$P(\text{at least one occurrence of event } A) = 1 - P(\text{no occurrences of event } A)$$

Example: Manufacturing (1 of 4)

A factory of the Global Manufacturing Company was manufacturing products with a defect rate of 15% (based on data from the *Harvard Business Review*). If a customer purchases 12 of the products, what is the probability of getting at least one that is defective?

Example: Manufacturing (2 of 4)

Solution

Step 1: Let A = at least 1 of the 12 products is defective.

Step 2: Identify the event that is the complement of A .

$\bar{A}$ = **not** getting at least 1 defect among the 12 items
= all 12 items are good with no defects

Example: Manufacturing (3 of 4)

Solution

Step 3: Find $P(\bar{A})$.

$$P(\bar{A}) = P(\text{all 12 items are good})$$

$$= 0.85 \cdot 0.85 \cdot 0.85 \cdot 0.85 \cdot 0.85 \cdot 0.85 \cdot 0.85 \cdot 0.85 \\ \cdot 0.85 \cdot 0.85 \cdot 0.85 \cdot 0.85$$

$$= 0.85^{12} = 0.142$$

Step 4: Find $P(A)$ by evaluating $1 - P(\bar{A})$.

$$P(A) = 1 - P(\bar{A}) = 1 - 0.142 = 0.858$$

Example: Manufacturing (4 of 4)

For a group of 12 products, there is a 0.858 probability of getting at least 1 that is defective.

As quality control goes, this is too high.

The company needs to improve its manufacturing process so that the defect rate is lowered.

Conditional Probability (1 of 2)

- Conditional Probability
 - A **conditional probability** of an event is a probability obtained with the additional information that some other event has already occurred.

Conditional Probability (2 of 2)

Notation

$P(B | A)$ denotes the conditional probability of event B occurring, given that event A has already occurred.

Intuitive Approach for Finding $P(B | A)$

The conditional probability of B occurring given that A has occurred can be found by **assuming that event A has occurred** and then calculating the probability that event B will occur.

Formal Approach for Finding $P(B | A)$

The probability $P(B | A)$ can be found by dividing the probability of events A and B both occurring by the probability of event A :

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

Example: Pre-Employment Drug Screening (1 of 11)

Find the following using the table:

a. If 1 of the 555 test subjects is randomly selected, find the probability that the subject had a positive test result, given that the subject actually uses drugs. That is, find $P(\text{positive test result} \mid \text{subject uses drugs})$.

	Positive Test Result (Test shows drug use.)	Negative Test Result (Test shows no drug use.)
Subject Uses Drugs	45 (True Positive)	5 (False Negative)
Subject Does Not Use Drugs	25 (False Positive)	480 (True Negative)

Example: Pre-Employment Drug Screening (2 of 11)

Find the following using the table:

b. If 1 of the 555 test subjects is randomly selected, find the probability that the subject actually uses drugs, given that he or she had a positive test result. That is, find $P(\text{subject uses drugs} \mid \text{positive test result})$.

	Positive Test Result (Test shows drug use.)	Negative Test Result (Test shows no drug use.)
Subject Uses Drugs	45 (True Positive)	5 (False Negative)
Subject Does Not Use Drugs	25 (False Positive)	480 (True Negative)

Example: Pre-Employment Drug Screening (3 of 11)

Solution

a. **Intuitive Approach:** We want $P(\text{positive test result} \mid \text{subject uses drugs})$. If we assume that the selected subject actually uses drugs, we are dealing only with the 50 subjects in the first row of the table. Among those 50 subjects, 45 had positive test results, so we get this result:

$$P(\text{positive test result} \mid \text{subject uses drugs})$$

$$= \frac{45}{50} = 0.900$$

Example: Pre-Employment Drug Screening (4 of 11)

Solution

Formal Approach: The same result can be found by using the formula for $P(B | A)$ given with the formal approach.

$P(B | A) = P(\text{positive test result} | \text{subject uses drugs})$
where B = positive test result and A = subject uses drugs.

Example: Pre-Employment Drug Screening (5 of 11)

Solution

Use $P(\text{subject uses drugs and had a positive test result})$
 $= 45/555$

and $P(\text{subject uses drugs}) = 50/555$

to get the following results:

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

Example: Pre-Employment Drug Screening (6 of 11)

Solution

becomes

$P(\text{positive test result} \mid \text{subject uses drugs})$

$$= \frac{P(\text{subject uses drugs and had a positive test result})}{P(\text{subject uses drugs})}$$

$$= \frac{\frac{45}{555}}{\frac{50}{555}} = 0.900$$

Example: Pre-Employment Drug Screening (7 of 11)

Solution

By comparing the intuitive approach to the formal approach, it should be clear that the intuitive approach is much easier to use, and it is also less likely to result in errors.

The intuitive approach is based on an *understanding* of conditional probability, instead of manipulation of a formula, and understanding is so much better.

Example: Pre-Employment Drug Screening (8 of 11)

Solution

b. Here we want $P(\text{subject uses drugs} \mid \text{positive test result})$. This is the probability that the selected subject uses drugs, **given that the subject had a positive test result**. If we assume that the subject had a positive test result, we are dealing with the 70 subjects in the first column of the table.

Example: Pre-Employment Drug Screening (9 of 11)

Solution

Among those 70 subjects, 45 use drugs, so

$P(\text{subject uses drugs} \mid \text{positive test result})$

$$= \frac{45}{70} = 0.643$$

Example: Pre-Employment Drug Screening (10 of 11)

Interpretation

The first result of $P(\text{positive test result} \mid \text{subject uses drugs}) = 0.900$ indicates that a subject who uses drugs has a 0.900 probability of getting a positive test result.

Example: Pre-Employment Drug Screening (11 of 11)

Interpretation

The second result of $P(\text{subject uses drugs} \mid \text{positive test result}) = 0.643$ indicates that for a subject who gets a positive test result, there is a 0.643 probability that this subject actually uses drugs. Note that $P(\text{positive test result} \mid \text{subject uses drugs}) \neq P(\text{subject uses drugs} \mid \text{positive test result})$.

This shows in general, $P(B|A) \neq P(A|B)$.

Confusion of the Inverse

In the prior example, $P(\text{positive test result} \mid \text{subject uses drugs}) \neq P(\text{subject uses drugs} \mid \text{positive test result})$. This example proves that in general, $P(B \mid A) \neq P(A \mid B)$. (There could be individual cases where $P(A \mid B)$ and $P(B \mid A)$ are equal, but they are generally not equal.)

Confusion of the inverse is to incorrectly think that $P(B \mid A)$ and $P(A \mid B)$ are equal or to incorrectly use one value in place of the other.

Example: Confusion of the Inverse (1 of 2)

Consider these events:

D: It is dark outdoors.

M: It is midnight.

Example: Confusion of the Inverse (2 of 2)

Solution

$P(D | M) = 1$ (It is certain to be dark given that it is midnight.)

$P(M | D) = 0$ (The probability that it is exactly midnight given that it dark is almost zero.)

Here, $P(D | M) \neq P(M | D)$.

Confusion of the inverse occurs when we incorrectly switch those probability values or think that they are equal.